



Various Legal Factors Extraction Based on Machine Reading Comprehension

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Abstract. With the rapid growth of legal cases, professionals are under pressure to go through lengthy documents and grasp informative pieces of text in the limited time. Most of the existing techniques focus on simple legal information retrieval task, such as name or address of the prosecutor or the defendant, which can be easily accomplished with the help of handcrafted patterns or sequence labeling methods. Yet complicated texts always challenge such pattern-based methods and sequence labeling approaches. These texts state the same facts or describe the same events, but they do not share common or similar patterns. In this paper, we design a unified framework to extract legal information in various formats, including directly extracted information (a piece of span) and information that needs to be deduced. The framework follows the methodology to answer questions in machine reading comprehension (MRC) tasks. We treat the extraction fact labels as the counterpart of questions in MRC task and propose several strategies to represent them. We construct several datasets regarding different cases for training and testing. Our best strategy achieves up to 4% enhancement in F1 score on each dataset compared to the MRC baseline.

Keywords: Information extraction · Machine reading comprehension · Legal information

1 Introduction

Nowadays, driven by increasingly complicated legal provisions and cases, both ordinary parties and legal workers are eager to use technical means to assist in analysis. In the process of assisting judicial work, it is an indispensable ability to extract various forms of required information efficiently and correctly. The information are sometimes pieces of text in the document, such as event descriptions, actions and entity names, or conclusions that are not directly stated and need deducing from the original text. Such information are usually crucial to the final sentence, which are called *legal factors* in the legal industry. Legal factors are closely cohered with legal case types and each type has a fixed factor list. Given the case type, judges or other legal workers are clear of what information to seek

from the document. However, different forms of legal factors cannot be easily extracted by a single model simultaneously using existing techniques. Thus, we hope to build a unified framework to extract all factors regarding the case type.

Generally, information extraction task [6] extracts entities (person, organization, etc.) and facts (relations, events, etc.) from given texts [16], helping to acquire the desired information and reconstruct massive contents. This is usually done by sequence labeling models, such as Lattice LSTM [18] and transformer-based models [9, 17]. They conduct experiments on Chinese named entity recognition (NER) task, and achieve high performance in benchmarks, such as Chinese NER MSRA [8], OntoNotes 4.0, and Resume NER [18]. However, they fail to maintain competitive results in our task. Our extraction targets are of multiple forms, including spans and deductions. Spans are often entity names and event descriptions, while deductions include answers to predefined legal conditions, for example, whether the document agrees (answers “Yes”) or opposes (answers “No”) to a given condition. These challenge the sequence labeling models since the answers are not spans from the context. Inspired by MRC tasks [7, 12, 13], whose question types are similar to ours, for example, asking for a named entity (span) and deducing a piece of opinion (“Yes” or “No”), we can employ such ideas to solve the legal factor extraction task. The difference lies in that the questions of MRC tasks are arbitrary, but the extraction fact labels (the counterpart to questions in MRC) are fixed along with the case type. We pre-define these fixed legal factors to a given case as a set of *extraction fact labels*.

In this paper, we design a unified framework, particularly for multi-formed legal factor extraction. It follows the MRC methodology to encode and interact with the document and the extraction fact labels. Some researchers proved that more specified and informative queries could improve the extraction precision [10]. As our extraction fact labels are fixed, to achieve better performance, we expand the labels to obtain more specified expressions with query expansion components before the extraction. Our framework solves the problem discussed above and our best strategy achieves 4% improvement compared to the baseline. Our contributions are as follows: i. We achieve automatic extraction of various forms of legal factor with a single model. ii. We proposed a strategy to represent legal factors that outperform the others.

2 Related Works

Some existing works investigate the feasibility of using MRC approaches to solving information extraction tasks. Li et al. [10] use the MRC model to solve the named entity recognition task and achieved good results in nested entity recognition. For each more nested entity, one more question requires to be answered. And the reading comprehension model is designed to handle the question-answering task. They suppose that the reading comprehension model is a natural solution to the nested entity problem. Similarly, in our task, information of factors is often nested or overlapped, we could exploit the MRC framework for legal factor extraction.

Query expansion is comprehensively used because of its simplicity and practicality. There are several ways to reach query expansion. The IBM algorithm in

the machine translation model is migrated to directly rewrite queries [5]. People find that it is too rough to directly rewrite queries, and instead use the Seq2Seq model to incorporate richer semantic information, and reinforcement learning is used to fine-tune the rewrite model [1]. Of course, there are other ideas, such as using a large amount of query click data, various concept words are mined out. Further, associating the concept words with the knowledge graph in order to replace the simple query clustering scheme [11].

3 The Legal Factor Extraction Framework

3.1 Architecture of the Framework

Primarily, we introduce two important concepts, *legal factors* and the *legal factor extraction* task. As mentioned in Sect. 1, legal factor is a jurisprudence concept, referring to the key information that affects the final sentencing, which includes the descriptions of a judicial event, such as persons, actions, causes, consequences and etc., and the conclusions to predefined judicial conditions. We define the legal factor extraction task as the procedure to find out the text stating such information from original paragraphs. Our framework consists of three parts: input documents and extraction fact labels embeddings, documents and labels interactive encoding, and result answers prediction. The overall structure is shown in Fig. 1 and Fig. 2 is an example shows how our framework actually works.

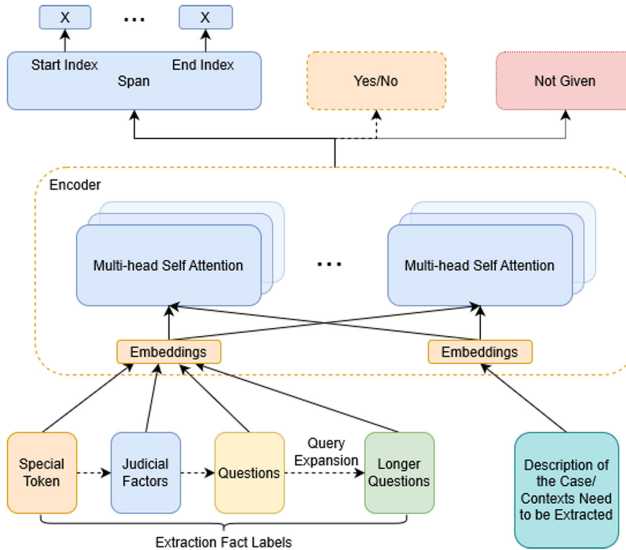


Fig. 1. Architecture of the framework. All the four forms of extraction fact labels can be fed into the encoder independently for extraction. The dotted arrows refer to the semantic enhancement.

In the embedding part, the inputs are the case descriptions and extraction targets. Note that the case type is an inherent attribute of a case description and the targets of extraction are determined along with the case type. We design different representations of the extraction fact labels and gradually enriched their semantic. First, we employ special tokens ([Token1], [Token2], etc.) as the targets. These tokens function as signals prompting the model to extracting different legal factors. Then, we represent the labels by text of legal factors, such as “伤情等级” (the injury level) and “经过/手段” (course/means). Further, the legal factors are expanded into more specific questions, for instance, “被害人伤情等级?” (What is the victim’s injury level?) and “被告人是否使用工具?” (If the defendant use tools?). Finally, the questions are reinforced to form longer questions via query expansion methods. We will discuss different extraction targets in the following subsection. The rest parts interactively encode documents and extraction fact labels with multi-head self attention [15], and predict the results according to the types of extraction fact labels. If it is unanswerable, an empty string returned. Otherwise, it answers “Yes” or “No” to a yes/no question, or returns the result of the extraction through the start and end indexes in the input passage.

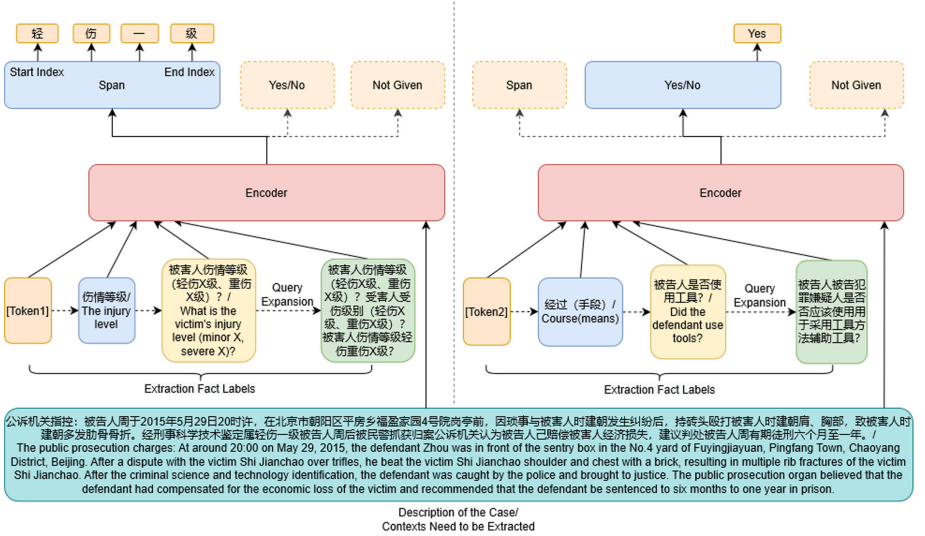


Fig. 2. Examples of the workflow (Left: Answer as span. Right: Answer as “YES”). We do not translate sentences after data augmentation (DA) because DA is usually a paraphrase of the original question, and the English version could remain unchanged.

3.2 Strategies for Representing Extraction Fact Labels

The Special Tokens. To verify our hypothesis that the different forms of legal factor can be extracted by a unified framework, we represent each extraction fact label as different [Token] rather than using the actual text of factor. They

act as signals guiding the model to extract designated content. Specific, we set a series of special tokens according to the legal factors that need to be extracted, where the factors are one-to-one mapped to the tokens. It is equivalent to cipher or semaphore. When the model encounters a special token, it learns to extract a specific kind of information. Commencing from the special tokens, we enrich the semantic gradually in the rest extraction fact labels representations.

From Factor to Question. Our produced datasets are dedicated to the type of cases, and the information that needs to be extracted from each dataset is summarized according to the experiences of the judiciary (legal factors). Taking the dataset of traffic accidents as an example, the factors are “事故时间” (time of the accident), “事故地点” (site of the accident), “责任划分” (division of responsibilities), “造成的损失” (the loss), etc. These factors are the counterpart of questions in typical question-answering datasets such as SQuAD [13], SQuAD2.0 [12], and CJRC [4]. Although the factors are not in the form of interrogative sentences, they contain all the core information needed for extraction. Hence, we use it the factors as another form of extraction fact labels.

Nevertheless, the factors are not written in general question format, they are just short key phrases indicating what content to look into. We move forward to gain question-like text to see if expanding a factor into question format benefits the extraction. We convert the factors into question format according to linguistic rules such as the wh-question patterns. For instance, “事故时间” (time of the accident) is a factor in the dataset of traffic accidents. We apply linguistic rules to switch it to its corresponding question, which results in “事故发生的时间是什么?” (What was the time of the accident?). Similarly, “事故地点” (site of the accident) is switched to “事故发生的地点是什么?” (Where did the accident take place?), and “造成的损失” (the loss) turns to “原告方遭受的损失有哪些?” (What losses does the plaintiff suffer?). Some factors are expanded into more than one question because they refer to multiple concrete facets. For example, “责任划分” (division of responsibilities) becomes “被告方的责任是什么?” (What are the responsibilities of the defendant?) and “原告方的责任是什么?” (What are the responsibilities of the plaintiff?). Such expansion literally enriches the semantic compared to the vanilla factors, and is highly likely to enhance the performance in the question-answering experiment.

From Question to Questions. Following the outcome (converted question) of last section, we use query expansion to further enrich the semantic information of the extraction factor labels. In this section, each extraction factor label is represented by longer or multiple questions. The expansion is quite simple and straightforward, namely synonym expansion. Specifically, there are two ways to expand a sentence with its synonym:

- The first is word-wise synonym expansion. A sentence is first segmented into words using LTP [2] as the word segmentation module. Then the corresponding synonyms of each word are directly concatenated following the original

- word order. We design two strategies to find the synonyms of each word, “ M_1 ” and “ M_2 ”. “ M_1 ” looks up the existing dictionary for synonyms, such as BigCilin [19]. While “ M_2 ” builds a special dictionary concerning both contexts and questions in each dataset. The embeddings of words in the whole created dictionary are generated by BERT [3]. We use the embedding vectors for calculating cosine similarities of the input word and the rest words from the dictionary. We rank the candidates and resume top K words as retrieved synonyms. Then, the synonyms are directly piled up to form a new sentence. For example, in the dataset of recourse for labor remuneration when supposing $K = 3$, the question “当事人是否签订劳动合同?” (Whether the party signs the labor contract?) will be expanded as “当事人原告被告 是否应该 签订签定签署 劳动劳动者劳动力 合同合约合同规定?”.
- The second is sentence-wise synonym expansion and we name it data augmentation (DA). The sentence is treated as a whole when searching for synonyms. To augmentate a whole sentence, we need to gain its the general semantic representation. Hence, we rely on the corpus of Baidu zhidao, which is a knowledge encyclopedia written in Chinese. This corpus is used to pre-train language model that better understand the correlation between Chinese words and phrases. We refer readers to SimBERT [14] for training details. We use this model to gain a ranked list of K similar sentences. After acquiring the sentence-wise synonyms, we propose several strategies to join them. In addition to direct concatenation, we utilize bi-LSTM and bi-GRU to encode semantic of these synonyms. Compared to the direct concatenation, this strategy extract hidden features and control dimension of the output in order to satisfy BERT’s input length limit. The direct concatenation easily exceeds this limitation and the rest part of text is ignored. We report selected augmentation results of $K = 5$ in Table 1 and Table 5 (in the appendix).

Table 1. Examples of data augmentation of recourse for labor remuneration (RLR) (See Appendix for other datasets). We report two typical questions for each dataset. Those without translations share the same English version with the original question’s.

Cases	Original Question	Data Augmentation (DA) (K=5)
RLR	劳动者的基本工资数额是多少? (What is the basic salary of workers?)	劳动者工资的基本数额是多少?
		劳动者的基本工资是多少?
		劳动者每月工资基数是多少?
		劳动者每月基本工资是多少?
		劳动者工资的基本数额和年终奖是多少? (How much are the basic salary and bonus?)
	离职的原因是什么? (What’s the reason for quitting?)	离职的主要原因是什么?
		离职是什么原因造成的? (What causes the quitting?)
		辞职的原因是什么?
		员工离职的原因是什么?
		辞职的原因是什么?

4 Experiments

4.1 Datasets and Experimental Settings

We composed three datasets regarding the following case types, intentional injury (II), recourse for labor remuneration (RLR), and refusal to execute judgments or rulings (REJR). The statistics are shown in Table 2. We select 200 question-answer pairs from each dataset as the testset. Note that the number of questions answered by “No” is markedly low. This is a common feature of judicial data, that officers tend to ask what they’ve already had evidence by hands, leading to few negative answers. We do not manually disturb this, because we believe that it is better to restore the actual dataset distribution.

As for experimental settings, we employ BERT-base encode the documents and extraction fact labels and refer readers to BERT [3] for detailed model descriptions and hyper-parameter settings. Since the input length of Bert is limited by 512, the length (L_F) allocated to the extraction fact label should also be constrained and well-designed. Overall, the input extraction fact label tokens of our model is no longer than 60, and the rest are preserved for document input. Specific, $L_F = 1$ if using [Token] as the representation; $L_F \leq 10$ for factors and questions; while $L_F \leq 60$ for expansions. In practice, most of the expanded questions are longer than 10, so any $K > 5$ results in input length suppressing. We will seek the best effect by varying K .

Table 2. Statistics of datasets. # refers to the number. NG refers to “Not Given”.

Cases	#Factors	#Document	#Questions	Answer types			
				Yes	No	Span	NG
RLR	17	992	5628	1233	189	8422	946
II	13	974	9840	695	246	7840	1059
REJR	13	960	8320	2017	25	4142	2158

4.2 Experiments of Different Methods for Legal Factor Extraction

Before carrying out experiments with MRC models, we try the sequence labeling methods to address this task. We convert our legal factor extraction datasets into the sequence labeling format, where the case descriptions in our datasets are labeled with BIOES (B-begin, I-inside, O-outside, E-end, S-single) tags. Further, we reproduce two entity recognition algorithms, Lattice LSTM [18] and Chinese BERT-base [3]. Because these models require to find span text from the given passage, only questions answered by span are preserved and the corresponding questions turn out to be the tags. The number of such tags are limited, so that we could number as Q1, Q2, etc. Specifically, for the question Q1, we find the corresponding answer span in the context and label the tokens within the span as “B-Q1 I-Q1 I-Q1 ... E-Q1”. The sequence labeling experiments are compared to our proposed question-answering approaches in Table 3. Unfortunately, the results do not live up to expectations, which we will discuss in Sect. 5.

4.3 The Structure of Questions and the Design of Query Expansion

We investigate different methods to figure out the query expansion module that works the best and design a series of ablation studies. In this section, we introduce these methods in details:

- “ Q_{ori} ”: To Directly generate questions from the factors. The result of the question-answering experiment in “ Q_{ori} ” will be the baseline of the whole experiments.
- “*modify*”: We manually modify some questions, mainly to fix some mistakes caused by the generation process and to make them closer to the contexts. The modification includes correction of typos and examination of punctuation characters. In the II dataset, we modify the question “受害人的受害经过是怎样的?” (What happened to the victim?) to “被害人的被害经过是怎样的?”. Although the English translation for both “受害人” and “被害人” is victim, only “被害人” appears in the contexts of the dataset rather than “受害人”. Therefore, we modify this word to be consistent with the context.
- “ $+F$ ”: To directly concatenate the factors with original questions as a whole.
- “ QE_{syn-M1} ”: Word-wise expansion using method $M1$ described in Sect. 3.2.
- “ QE_{syn-M2} ”: Word-wise expansion using method $M2$ described in Sect. 3.2.
- “ $QE_{da-random}$ ”: To replace the original question with a randomly question obtained by data augmentation.
- “ $QE_{da-lstm}$ ”: To add a bi-LSTM layer after data augmentation to extract semantic information from augmented questions.
- “ QE_{da-gru} ”: To add a bi-GRU layer after data augmentation to extract semantic information from augmented questions.
- “ $QE_{da-topK}$ ”: To directly concatenate the first K augmented sentences with the original question. We experiment with $K = 1, 3, 5$.
- “ QE_{da-mix} ”: To only expand questions whose answers are spans, but use the original question for those answered by “Yes” or “No”, and those with no answers (NG).
- “ QE_{da-yn} ”: Do not expand question those answered by “Yes” or “No”, but perform expansion on other kinds of questions.
- “ $QE_{da-last5}$ ”: To concatenate the last five out of ten augmented sentences with the original question. We hypothesize that taking the last five sentences increases the diversity of semantic because they are less similar to the original question than the first five sentences.

Among, the above strategies, $QE_{da-topK}$, QE_{da-mix} , QE_{da-yn} and $QE_{da-last5}$ are four ablation studies that reveal what indeed leads to the enhancement of extraction results.

4.4 Results

The results of different methods to extract legal information and the performances of different forms of extraction fact labels are reported in Table 3. Not as

expected, the sequence labeling methods much underperform the rest methods. The F1 score of Lattice only reaches 11.17%. Similarly, *Seq2SeqBERT* does not perform much better, with F1 score 22.8%. We can conclude that the sequence labeling methods are incapable of handling our task. Among the forms of extraction fact labels, the one with query expansion module achieves best F1 results for RLR (88.51), II (86.11) and REJE (81.47) datasets. This method also contributes to the highest EM scores on RLR (77.07) and REJR (67.34) datasets. While in the II dataset, representing fact labels as questions peaks by the EM score of 74.88. Table 3 shows that along with the increasing of semantic information from tokens to expanded questions, the results are broadly improved, reaching up to 20% in F1 and 15% in EM. These prove the idea that richer semantic features boosts the model’s performance.

Table 4 and Table 6 (in the appendix) report the results of query expansion strategies and the ablation experiments. Taking RLR dataset as an example, we apply all the methods and strategies described in Sect. 4.3 to test the effectiveness. Among these strategies, the data augmentation-based strategies outperform the others. The synonym-based experiments show similar pattern as the RLR for the other two datasets. So we omit the synonym-based results due to the page limitation and focus on the data augmentation-based experiments. In RLR dataset, “*QEda-top5*” has a positive effect and its EM and F1 enhance about 1.5% and 2%, respectively compared to “*QEda-lstm*” and “*QEda-gru*”. After these experiments, we realize that query expansion methods work unequally for all kinds of questions. Some work well on questions whose answers are spans, but poorly on “Yes”, “No”, or “NG” questions. Hence, apart from expanding question of all answer types (*QEda-topK*), we design another experiments, *QEda-mix*, *QEda-yn*, and *QEda-last5*, as ablation experiments to test the influence on different types of extractions and to find out the best strategy. The *QEda-mix* achieves an overall good result and is the recorded strategy in Table 3, but it affects datasets differently.

Table 3. Exact Match(%) and F1(%) of different forms of extraction fact labels (token, factor, question and expanded question) and sequence labeling method. Specific, $F1_{Seq1}$: the F1 of Lattice [18]; $F1_{Seq2}$: the F1 of *Seq2SeqBERT*; Expansion: *QEda-mix*.

Cases	Seq. labeling		Token		Factor		Question		Expansion	
	$F1_{Seq1}$	$F1_{Seq2}$	EM	F1	EM	F1	EM	F1	EM	F1
RLR	11.17	22.80	69.39	82.04	72.96	85.49	73.98	85.13	77.04	88.51
II	–	–	69.35	81.30	71.36	82.92	74.88	86.10	72.86	86.21
REJR	–	–	52.26	60.66	58.29	66.24	65.83	80.76	67.34	81.47

5 Analyses

5.1 Extract Legal Factor as a Human Judge

Table 3 examines the feasibility of sequence labeling approaches in solving the proposed legal factor extraction task and confirms that they are unsuitable to our task. The “entities” in our datasets are nearly ten times as long as the entities

Table 4. Results of special questions, reported in **Exact Match**(%), **F1** score(%), **Precision**(%) and **Recall**(%). Details of these methods are shown in Sect. 4.4

Cases	Methods	Total		Span		Not given		
		EM	F1	EM	F1	R	P	F1
Recourse for Labor Remuneration (RLR)	Q_{ori}	72.96	84.40	63.21	84.36	80.00	82.35	81.16
	$Q_{ori} + F$	70.92	83.98	57.55	81.70	82.86	80.56	81.69
	$Q_{ori-modify}$	73.98	85.38	61.32	82.40	85.71	83.33	84.50
	QE_{syn-M_1}	70.92	82.88	62.26	84.38	68.57	85.71	76.19
	QE_{syn-M_2}	71.94	84.36	59.43	82.39	80.00	80.00	80.00
	$QE_{syn-M_2} + F$	72.96	84.01	63.21	83.64	77.14	79.41	78.26
	$QE_{da-random}$	70.41	82.72	59.43	82.21	74.29	81.25	77.61
	$QE_{da-top5}$	75.51	86.25	66.98	86.84	80.00	87.50	83.58
	$QE_{da-lstm}$	71.43	84.39	58.49	82.46	82.86	78.38	80.56
	QE_{da-gru}	71.43	83.59	58.49	80.97	82.86	82.86	82.86
	QE_{da-mix}	77.04	88.51	63.21	84.41	100.00	92.11	95.89
	QE_{da-yn}	71.43	83.38	58.49	80.59	82.86	76.32	79.46
	$QE_{da-top3}$	71.43	82.91	59.43	80.66	82.86	74.36	78.38
	$QE_{da-top1}$	73.47	84.70	63.21	83.97	80.00	80.00	80.00
	$QE_{da-last5}$	72.96	84.73	62.26	84.04	80.00	82.35	81.16
Intentional Injury (II)	Q_{ori}	72.36	84.78	68.26	83.06	90.00	85.71	87.80
	$Q_{ori-modify}$	72.36	85.08	68.26	83.42	90.00	85.71	87.80
	$QE_{da-top5}$	71.86	83.56	68.26	82.21	85.00	89.47	87.18
	$QE_{da-lstm}$	70.85	83.70	67.66	82.98	80.00	84.21	82.05
	QE_{da-gru}	70.35	82.68	65.87	80.56	90.00	85.71	87.80
	QE_{da-mix}	72.86	86.21	69.46	85.36	85.00	89.47	87.18
Refusal to Execute Judgments or Rulings (REJR)	Q_{ori}	66.33	80.23	50.00	77.11	75.00	82.98	78.79
	$Q_{ori-modify}$	67.34	81.22	50.00	77.09	78.85	87.23	82.83
	$QE_{da-top5}$	67.34	81.19	50.98	78.00	76.92	86.96	81.63
	$QE_{da-lstm}$	65.83	80.01	49.02	76.69	78.85	82.00	80.39
	QE_{da-gru}	67.34	80.67	51.96	77.96	75.00	88.64	81.25
	QE_{da-mix}	66.83	81.47	49.02	77.57	78.85	91.11	84.54

in the standard NER datasets, such as Resume NER [18]. Our “entities” are essentially the answer spans to pre-defined questions, and are uneven in length. Whilst in the other datasets, most of the entities are of just a few tokens, and the number of entity tags are far less than ours. Therefore, the sequence labeling method is unqualified for information extraction task of this magnitude. These urges us to introduce the MRC-based means as better solutions.

In general MRC task, the model knows what answers to searching for because the questions are written in plain text. But in the “Token” experiment, we only feed the token symbols to the model without the specific target of the extraction descriptions. By doing so, we imitate the procedure of a judge reading through the documents. As we discussed in Sect. 1, these legal factors are fixed regarding the case type, and a judge can fast decide the what to looking for when

given a case description. This experiment proves that the model can learn the semantic meanings through this special token without been told the exact target. To a certain extent, it completes the work of expert knowledge and artificial settings by itself. This differs our methods from general MRC approaches, and is instructive for follow-up experiments.

5.2 The Richer Semantic Information, the Higher Score

We can conclude from Table 3 that along with the increasing of semantic information from tokens to expanded questions, the performances are improved remarkably. Starting from the Token representation to the Factor and Question, and eventually the Expansion, the semantic meaning of the extraction fact labels get more specific and enhanced. Their corresponding results also get boosted gradually. In the most extreme example, the REJR dataset, the F1 score increases from about 60 to over 81, and the EM score grows about 15 from Token to Expansion. As shown in Table 3, there are surely information growth from Token to Factor and from Question to Expansion. However, in the RLR dataset, the trend from Factor to Question is in contrast with the others. A possible explanation is that the Factor description is already concrete enough and converting it into question format introduces undesired noises. Overall, the expressiveness of different forms of extraction fact labels get enhanced all the way from Token to Expansion. This consequently achieves higher scores.

5.3 Strategies of Query Expansion

Following the results, we further discuss different types of query expansion and the corresponding detailed strategies. Table 4 proves that data augmentation-based approaches outperform those synonym-based. Unexpectedly, synonym-based approaches sometimes impair the performance. Different from synonym conversion, the augmentation adds extra meaningful words to the questions. For instance in the RLR dataset, the question “劳动者的基本工资数额是多少?” (What is the basic salary of workers?) turns out to be “劳动者工资的基本数额是多少?”, “劳动者的基本工资是多少?” and “劳动者每月工资基数是多少?”, where $K = 3$. Although the subject of the sentence, “劳动者” (workers), does not change to its synonyms, yet the object “基本工资数额” (the amount of basic salary) becomes “工资的基本数额” (the basic amount of salary), “基本工资” (basic wage) and “工资基数” (wage base). And in the third enhanced sentence, a new word “每月” (monthly) is added as the attributive of “工资基数” (wage base). Obviously, data augmentation brings richer semantic features to the original questions, including paraphrasing of the original words and introducing of extra information. These make the questions more accurate and specific.

Nevertheless, the more is not always the better. We discover that different augmentations have inconsistent effects on different datasets and different types of questions. Expanding the question for all the answer types (span, YES, NO and NG) does not always give the best results (see results of $QE_{da-topK}$ in

Table 4 and Appendix Table 6). In the ablation experiments, some questions are preserved and the others are expanded. We first expand all the questions to find out what types of questions end in poor results, and leave these questions not expanded while expand the rest questions who have outstanding performance. The final strategy is *QE_{da-mix}*, whose experimental results on all three datasets get improved by up to 4 points in both F1 and EM.

For different datasets, the same type of questions does not necessarily show similar results before and after query expansion. For example, in the RLR and REJR datasets, query expansion works very well on questions whose answers are spans, and poorly on questions whose answers are “YES”, “NO” and “NG”. But in the II dataset, the results are exactly the opposite. Practically, it is difficult for us to fully judge the results of the experiments through pre-speculation. For instance, we intuitively thought that it is unnecessary to expand yes/no questions because their answers are simple and naive, and the question format is uniform. However, as far as the experimental results are concerned, our speculation can only be incompletely wrong. The performance of different types of questions before and after query expansion could be inconsistent, and cannot be predicted despite of the datasets they belong to.

6 Conclusion

In this paper, we design a unified framework to extract different types of legal factors. It is a MRC-based framework but with pre-determined extraction fact labels according to the datasets and achieves automatic extraction without manually feeding the questions. To verify the effectiveness of our approach, we constructed three datasets, including intentional injury (II), recourse for labor remuneration (RLR), and refusal to execute judgments or rulings (REJR) cases. Experiments show that such model suits well to our information extraction task compared to the sequence labeling models. To improve the model performance, we design different strategies and conduct plentiful experiments that discuss the impact of adding semantic information to the extraction fact labels. The strategies fall into two main categories, synonym-based and data augmentation-based. Among all the strategies of these two categories, the *QE_{da-mix}* performs generally well, improving the performance of up to 4% on each dataset.

7 Future Works

Our proposed framework aim at extracting different types of legal factors. So far, it is able to extract span-like factors, deduce yes/no factors and judge whether a factor is included in the document or not. However, there are still several types of legal factors that are not supported, such as multiple-choice factors, numerical derivation factors, summarisation factors, etc. multiple-choice factors refer to the fact that involves a fixed number of parties. For instance, some disputes over rental contracts often involve three parties: the landlord, the tenant, and the intermediary. The numerical derivation factors usually ask for the total amount of money, weight, and the number of criminal activities. These information are

not directly and clearly written, nor can they be derived during straightforward extractions. In the future, we will manage to expand the framework’s capability to let it handle more types of factors.

Experiments show that although query expansion has a positive effect on each dataset, yet different datasets have different responses to query expansion. For the RLR and REJR datasets, query expansion works well on questions whose answers are spans. While query expansion gives a good result on the other types of instance for II dataset. We believe this is due to differences in the structures and content of different case types, but we have not quantified these differences or looked for common patterns. In the future, we will try to find a universal query expansion strategy to streamline the current solution, as well as adding new factor types into consideration.

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Appendix

Table 5. Data augmentation of other two datasets. We selected two representative questions from each dataset. We do not translate sentences after data augmentation (DA) because DA is usually a paraphrasing of the original questions, and the English version could remain unchanged.

Cases	Original Question	Data Augmentation(K=5)
II	被告人作案工具是什么？ (What is the defendant’s tool of committing the crime?)	什么是被告人的作案工具？
		被告人作案的工具都有什么？
		被告人作案工具？
		被告人作案时应该用什么工具？
		被告人作案工具有哪些？
	损失的经济构成？ (What is the economic composition of the loss?)	损失的经济构成？
		损失经济构成要点？
		损坏经济损失的构成要件？
		经济损失的构成要件有哪些？
		经济损失的构成要素是什么？
REJR	被告人义务内容？ (What are the obligations of the defendant?)	被告人义务的内容？
		被告人的义务是？
		被告人义务的内容包括哪些？
		被告人的义务有哪些？
		被告人的义务是什么？
	是否对被告人限制高消费？ (Is the defendant restricted to high consumption?)	被告人限制高消费是否有效？
		被告人是否能够限制高消费？
		法院对被告人高消费，是否有限制的？
		对被告人的高消费行为是否有规定？
		对被告人限制高消费的行为有哪些？

Table 6. Results of general questions, reported in **Exact Match**(%), **F1** score(%), **Precision**(%) and **Recall**(%). The meaning of each experiment in this table is the same as that in Table 4. It is noted that the extraction results of the questions whose answer is no in the datasets of recourse for labor remuneration and refusal to execute judgments or rulings are poor, and even the results of many experiments are 0. As shown in Table 2, the questions with negative answer are originally very rare, and we do not build the datasets specifically for this situation, which made it difficult to extract, or simply do not show in the test sets.

Cases	Methods	Yes			No		
		R	P	F1	R	P	F1
Recourse for Labor Remuneration (RLR)	Q_{ori}	97.96	92.31	95.05	0.00	0.00	0.00
	$Q_{ori} + F$	100.00	90.74	95.15	0.00	0.00	0.00
	$Q_{ori-modify}$	100.00	94.23	97.03	16.67	100.00	28.57
	QE_{syn-M_1}	100.00	83.05	90.74	0.00	0.00	0.00
	QE_{syn-M_2}	97.96	94.12	96.00	33.33	50.00	40.00
	$QE_{syn-M_2} + F$	100.00	87.50	93.33	0.00	0.00	0.00
	$QE_{da-random}$	100.00	89.09	94.23	0.00	0.00	0.00
	$QE_{da-top5}$	100.00	90.74	95.15	0.00	0.00	0.00
	$QE_{da-lstm}$	100.00	92.45	96.08	0.00	0.00	0.00
	QE_{da-gru}	100.00	92.45	96.08	0.00	0.00	0.00
	QE_{da-mix}	100.00	92.45	96.08	0.00	0.00	0.00
	QE_{da-yn}	100.00	92.45	96.08	0.00	0.00	0.00
	$QE_{da-top3}$	97.96	92.31	95.05	0.00	0.00	0.00
	$QE_{da-top1}$	100.00	90.74	95.15	0.00	0.00	0.00
	$QE_{da-last5}$	100.00	92.45	96.08	0.00	0.00	0.00
Intentional Injury (II)	Q_{ori}	100.00	90.00	94.74	100.00	100.00	100.00
	$Q_{ori-modify}$	100.00	90.00	94.74	100.00	100.00	100.00
	$QE_{da-top5}$	100.00	90.00	94.74	100.00	100.00	100.00
	$QE_{da-lstm}$	100.00	90.00	94.74	100.00	100.00	100.00
	QE_{da-gru}	100.00	100.00	100.00	100.00	100.00	100.00
	QE_{da-mix}	100.00	90.00	94.74	100.00	100.00	100.00
Refusal to Execute Judgments or Rulings (REJR)	Q_{ori}	95.45	76.36	84.84	0.00	0.00	0.00
	$Q_{ori-modify}$	95.45	77.78	85.71	0.00	0.00	0.00
	$QE_{da-top5}$	95.45	77.78	85.71	0.00	0.00	0.00
	$QE_{da-lstm}$	90.91	80.00	85.11	0.00	0.00	0.00
	QE_{da-gru}	95.45	77.78	85.71	0.00	0.00	0.00
	QE_{da-mix}	95.45	77.78	85.71	0.00	0.00	0.00

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